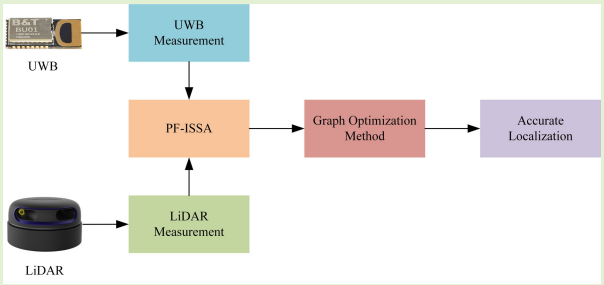
UWB/LiDAR Tightly Coupled Positioning Algorithm Based on ISSA Optimized Particle Filter

Xuyin Wang, Fangzheng Gao, *Member, IEEE*, Jiacai Huang, *Member, IEEE*, and Yuan Xue

***Abstract*—To deal with the disadvantages of non-line-of-sight (NLOS) errors in ultrawideband (UWB) and cumulative errors in LiDAR which impact positioning accuracy, a UWB/LiDAR tightly coupled positioning method is presented in this article by the improved sparrow search algorithm (ISSA) optimized particle filter. By incorporating LiDAR measurements, this method offers the distance estimation between the combined positioning system and the UWB base station and eliminates the NLOS errors in the UWB measurement value. In addition, the ISSA is used to eliminate the particle degradation phenomenon of particle filter and reduce the required number of particles, which significantly enhances the speed and real-time performance of the data fusion algorithm. Finally, the combined function of UWB/LiDAR is constructed to optimize the global position based on the graph optimization method. The experimental results demonstrate that the particle filter algorithm optimized by ISSA achieves comparable results while using only 25% of the particles required by the original particle filter algorithm. Moreover, the proposed method improves the positioning accuracy by 69.16% and 59.63% compared with UWB and LiDAR alone, and it improves the positioning accuracy by 55.71% compared with fusion positioning using extended Kalman filter (EKF). These results highlight the effectiveness of the proposed method in achieving accurate positioning.**



***Index Terms*— Graph optimization, improved sparrow search algorithm (ISSA), indoor position, particle filter.**

I. Introduction

With the continuous advancement of information technology, the positioning services of various devices have reached a level of maturity, and their applications are becoming increasingly widespread [1], [2]. In open outdoor environments, global navigation satellite system (GNSS) can provide highly precise, robust, and secure positioning [3]. However, in indoor environments, the GNSS signals are severely attenuated due to building obstructions, which renders it inadequate to meet the requirements of high-precision indoor positioning [4]. To solve such problem, some advanced indoor positioning technologies are currently developed, which include ultrawideband (UWB) [5], radio frequency identification (RFID) [6], LiDAR [7], ultrasonic, and inertial positioning technologies [8]. The accuracy ranges of these commonly used positioning technologies are illustrated in Fig. 1.

In comparison to other indoor positioning technologies, UWB positioning technology possesses the distinct advantages of high positioning accuracy, low device power consumption, strong signal penetration, and high time resolution. As a result, UWB is frequently used in the indoor scenarios that demand high-precision positioning [9]. UWB positioning technology is capable of delivering high-precision and reliable positioning information under line-of-sight (LOS) conditions. However, in the indoor environments characterized by complexity and numerous obstacles, UWB positioning devices often operate under the non-LOS (NLOS) conditions, which significantly degrades the accuracy of UWB positioning [10]. On the other hand, LiDAR positioning exhibits favorable spatial characteristics; however, it also suffers from certain drawbacks, such as the inability to quickly search for the positioning starting point and the accumulation of errors during long-term operation in large indoor environments.

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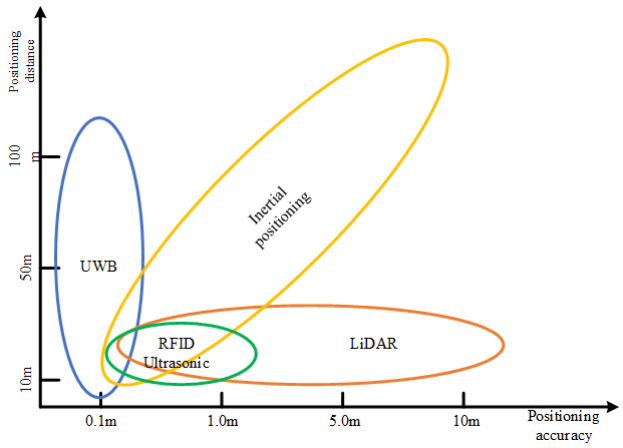


Fig. 1. Accuracy ranges of commonly used positioning technologies.

These inevitable factors lead to unsatisfactory positioning accuracy [11].

It is clear that single UWB or LiDAR sensors have limitations that prevent them from meeting the high accuracy and robust positioning requirements of modern complex applications. One of the effective solutions to such limitations is enhanced positioning capabilities by leveraging the strengths of different sensors. In recent years, a growing focus is made on exploring effective fusion techniques for UWB, LiDAR, and other multisensor systems [12]. Currently, the majority of the existing researching products mainly concentrate on the fusion of different sensor localization data. By combining UWB and LiDAR data, a positioning approach is proposed in [13] to mitigate the cumulative error associated with LiDAR and improve positioning accuracy of the system. To tackle the issue of inaccurate positioning in low-textured environments, Nguyen et al. [14] introduce a positioning method that uses data from inertial measurement units (IMUs), UWB, LiDAR, and vision sensors to enhance positioning accuracy in challenging environments. The fusion approach is specifically designed to leverage the synergistic information from these multiple sensors, aiming to improve the precision of positioning. In [15], the extended Kalman filter (EKF) algorithm is used to fuse UWB and LiDAR data, which capitalizes on the strengths of UWB's robust anti-interference capability and high time resolution, as well as LiDAR's efficiency and accuracy. In [16], a combination of LiDAR point cloud images and UWB anchor point information is presented by distinguishing between LOS and non-LOS (NLOS) data with real-time obstacle detection. Robust Kalman filter is used to mitigate the impact of NLOS errors, resulting in indoor positioning with subdecimeter-level accuracy. In [5], an improved loosely coupled positioning based on the Kalman filter is proposed to suppress the NLOS error of UWB and mitigates the drift of the IMU.

Note that the Kalman filter is an effective tool for multisensor fusion positioning in linear environments. However, the presence of obstacles renders that most of the indoor environments are nonlinear, where the particle filter demonstrates superior adaptability compared with EKF and unscented Kalman filter (UKF) [5]. Although the advantages are obvious, the particle filter is affected by a significant drawback known as particle degeneracy. This drawback leads to a wastage of computational resources due to the fact that the posterior probability only represents by a few particles with substantial weights while the majority of particles have little contribution [19]. Therefore, it is important and meaningful to explore how it overcomes the degradation phenomenon of particle filter and develops the novel improved particle-filter-based fusion positioning algorithm for high-precision indoor positioning in nonlinear environments. Nevertheless, to the best of our knowledge, the attention toward the study of this topic is not enough up to now.

Motivated by the above observations, this article aims to develop the tightly coupled fusion algorithm for high-precision indoor positioning in nonlinear environments by combining UWB and LiDAR simultaneous localization and mapping (SLAM) with the improved particle filter. The main innovations and contributions of this study are summarized as follows.

1) To fundamentally address the issue of particle degradation, the sparrow search algorithm (SSA) guidance mechanism is integrated into the structure of the particle filter algorithm. In this process, the mechanism of discarding low-weighted particles in the resampling stage is replaced with SSA's capability. As a result, the phenomenon of particle degradation is eliminated by the way that the spatial distribution of particles is optimized and managed throughout the algorithm's execution.
2) In view of the shortcomings of the original SSA [20], chaotic mapping is used to optimize the sparrow population distribution and improve its search strategy.
3) Different from the loosely coupled positioning [5], [15], [16], a novel tightly coupled fusion method is developed for indoor positioning with the help of the proposed improved SSA (ISSA) optimized particle filter. This method achieves a good result of accurate positioning by absorbing the respective advantages of UWB and LiDAR sensors and suppressing the shortcomings of NLOS errors of UWB and cumulative errors of LiDAR.

The rest of this article is structured as follows. In Section II, the positioning principles of UWB and LiDAR are offered, followed by an explanation of the SSA and its improvements. Then, the developed main algorithm process is presented. In Section III, the experimental platform and environment are briefly introduced, and then the experiments are conducted and analyzed. Section IV discusses the experimental results. Finally, this article is concluded in Section V.

II. Methods

This section briefly introduces the positioning principles of UWB and LiDAR sensors, and then gives the SSA and its improvements, and finally expounds the UWB/LiDAR indoor positioning method based on the ISSA optimized particle filter.

A. UWB Positioning Principle

Common UWB ranging methods include angle of arrival (AOA) [21], time of flight (TOF) [22], roundtrip TOF (RTOF) [23], and two-way TOF (TW-TOF) [24] methods,

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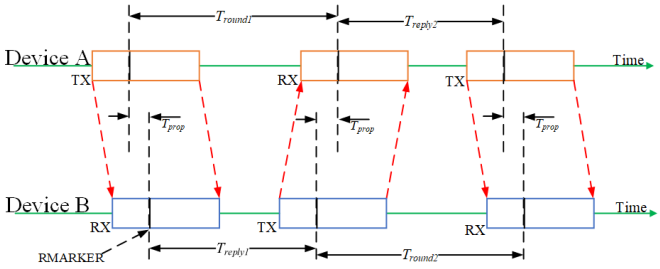


Fig. 2. TW-TOF ranging principle.

among others. The TW-TOF method is particularly noteworthy because it does not require communication time synchronization between the base station and the tag [25]. This method calculates the distance between the base station and the tag by measuring the time difference generated during data exchange, which can effectively eliminate the measurement error caused by clock accuracy [26]. The principle of the TW-TOF ranging method is illustrated in Fig. 2.

Based on the principle of the TW-TOF ranging method, the distance $d_{\text{TW-TOF}}$ between the base station and the tag is calculated as shown in the following equation:

$$d_{\text{TW-TOF}} = c\frac{T_{\text{round1}} \times T_{\text{round2}} - T_{\text{reply1}} \times T_{\text{reply2}}}{T_{\text{round1}} + T_{\text{round2}} + T_{\text{reply1}} + T_{\text{reply2}}} \tag{1}$$

where c is the approximate speed of electromagnetic wave in vacuum; T_{round1} and T_{round2} are the time difference between two communications of device A; and T_{reply1} and T_{reply2} are the time difference in two communications of device B.

In this article, the UWB positioning method of three base stations is adopted. That is, in the case that the positions of the three base stations are fixed and their distances are known, by measuring the distances d_1, d_2, and d_3 between the tags and each base station, one can obtain the position of the tags by the three-point positioning method. The principle of the three-point positioning method is shown in Fig. 3. The three base stations are, respectively, installed in three places indoors and keep the positions fixed with their positions described as $A_1(x_1, y_1, z_1)$, $A_2(x_2, y_2, z_2)$, and $A_3(x_3, y_3, z_3)$. The coordinates of the combined system are $T_0(x_0, y_0, z_0)$, with the height z_0 being known. Hence, the general formula for the relative distance between the base station and the tag is obtained as

$$r_i = \sqrt{d_i^2 - (z_i - z_0)^2}, \quad i = 1, 2, 3. \tag{2}$$

The main idea of the three-point positioning method [27] is that if a circular range is drawn based on the relative distance from each base station to the tag, then the intersection point of these three circular ranges is unique and definite. The coordinate (x_0, y_0) of the intersection point on this plane is calculated as follows:

$$\begin{cases} \sqrt{(x_0 - x_1)^2 + (y_0 - y_1)^2} = r_1 \\ \sqrt{(x_0 - x_2)^2 + (y_0 - y_2)^2} = r_2 \\ \sqrt{(x_0 - x_3)^2 + (y_0 - y_3)^2} = r_3. \end{cases} \tag{3}$$

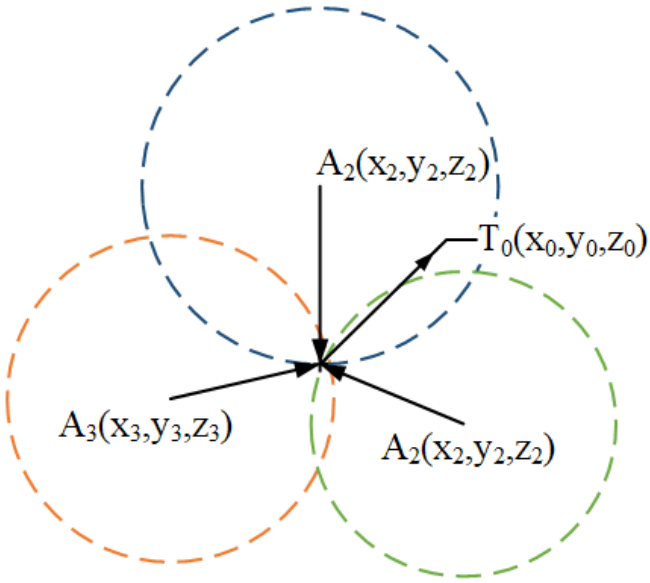


Fig. 3. Three-point positioning principle.

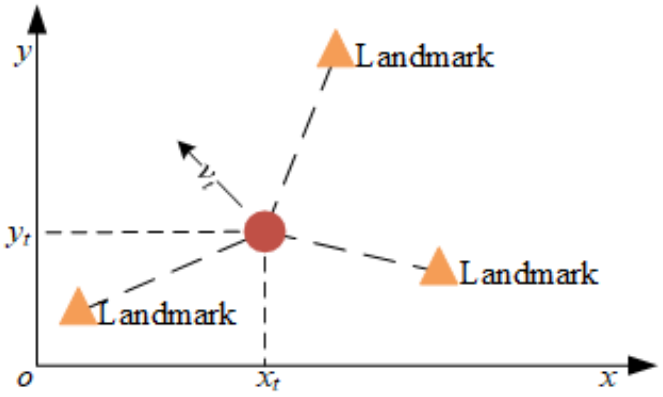


Fig. 4. LiDAR positioning principle.

In the actual positioning and ranging process, due to nonlinear factors such as indoor environment changes and obstacle occlusion, there are NLOS errors in UWB measurement information inevitably, which reduces the accuracy and stability of positioning. Therefore, LiDAR measurement data are introduced to offset the impact of NLOS errors.

B. LiDAR Positioning Principle

In indoor environments, there are fixed, unique, and easy-to-observe objects, which are called landmarks [28]. As shown in Fig. 4, by scanning these landmarks multiple times, LiDAR can estimate the position and attitude of its own environment.

At time t, the position of the landmark in LiDAR's own coordinate system and the position in the global coordinate system can be converted by the following equation:

$$x_t^L = T_t^{-1}(x_t^m - x_t) \tag{4}$$

where x_t^L is the coordinate obtained by observing the LiDAR's own coordinate system at time t; $T_t^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ is the transformation matrix between its own coordinate system and the global coordinate system; $x^m = [x^m, y^m]^T$ is the

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position of the landmark in the global coordinate system; and $x_n = [x_n \quad y_n]^T$ is the position of the LiDAR in the global coordinate system.

C. Sparrow Search Algorithm

Due to the presence of particle degradation in particle filter, it has become an effective way to enhance its performance by integrating the swarm optimization algorithm into the particle filter, which enables particle groups to concentrate in high-likelihood areas by simulating the activity of biological populations. Taking into account the SSA's advantages of strong optimization ability and high search accuracy [20], this article improves the particle filter by introducing the SSA algorithm into its resampling process. That said, a particle filter based on an ISSA (PF-ISSA) is developed in this article. In the PF-ISSA, "sparrows" are used to optimize the spatial distribution of particles rather than discarding low weight particles during resampling, and thus, particle degradation is fundamentally avoided.

The SSA is a bionic optimization algorithm inspired by sparrow foraging and antipredation behavior, and its fitness value matrix is as follows:

$$F_x = [f(x_1), f(x_2), \ldots, f(x_N)]^T \tag{5}$$

$$f(x_i) = [f(x_{i,1}), f(x_{i,2}), \ldots, f(x_{i,d})] \tag{6}$$

where N is the size of the sparrow population, and d is the dimension.

There are three parts in the sparrow population: discoverers, followers, and vigilantes. Individuals with high fitness in the population are called discoverers. Compared with other individuals in the population, discoverers have a larger search range, and their position update method is shown in the following equation:

$$X_{i,j}^{t+1} = \begin{cases} X_{i,j}^{t} \cdot \exp\left(-\dfrac{i}{\alpha \cdot \text{iter}}\right), & R_2 < ST \\ X_{i,j}^{k} + Q \cdot L, & R_2 \geq ST \end{cases} \tag{7}$$

where i represents the i-th sparrow in the population, $i = 1, 2, \ldots, n$; j represents the j-th dimension of the optimization problem, $j = 1, 2, \ldots, d$; t represents the t-th iteration of iterations of the algorithm; α is a random value in the interval $(0, 1]$; R_2 represents the early warning value in the process of foraging; ST represents the safety value in the process of foraging; Q is a random value following a normal distribution; and L represents a matrix of $1 \times D$. When $R_2 < ST$, the sparrow population is in a safe environment at this time, and the finder should continue to expand the search range. Otherwise, it means that there are natural enemies in the surrounding environment, and the population is in a dangerous area which should move to a safe area for food.

When the sparrow population is foraging, the follower in the population will follow the finder and its position is updated as follows:

$$X_{i,j}^{t+1} = \begin{cases} Q \cdot \exp\left(\dfrac{X_{\text{worst}} - X_{i,j}^{t}}{i^2}\right), & i > \dfrac{n}{2} \\ X_p^{t+1} + \left|X_{i,j}^{t} - X_p^{t+1}\right| \cdot A^{+} \cdot L, & i \leq \dfrac{n}{2} \end{cases} \tag{8}$$

where X_{worst} is the worst position of the sparrow in the population; X_p represents the optimal position reached by the discoverer in the population; and A represents a matrix of $1 \times D$, the elements of which are randomly 1 or -1, and $A^{+} = A^T(AA^T)^{-1}$. When $i > n/2$, it means that the follower did not grab the food that the discoverer was looking for and needs to go to other areas for foraging.

During foraging of the sparrow population, some sparrows will be selected as vigilantes to be responsible for the reconnaissance and early warning of the surrounding environment. When natural enemies appear in the surrounding environment, the vigilantes will give up the current food and approach other sparrows to avoid danger. The number of vigilantes is usually 10%–20% of the entire population, and the update method of their positions is as follows:

$$X_{i,j}^{t+1} = \begin{cases} X_{\text{best}}^{t} + \beta \cdot \left|X_{i,j}^{t} - X_{\text{best}}^{t}\right|, & f_i > f_g \\ X_{i,j}^{t} + k\dfrac{\left|X_{i,j}^{t} - X_{\text{worst}}^{t}\right|}{f_i - f_w + \varepsilon}, & f_i = f_g \end{cases} \tag{9}$$

where X_{best} represents the best position of the current sparrow population; β is a random value that obeys the normal distribution, which is used to control the moving step of the sparrow individual; k is a random number used to control the moving direction and distance of individual sparrows, and its range is $[-1, 1]$; ε represents the fitness value of the ith sparrow in the population; and f_g and f_w represent the fitness values of the global optimal and worst positions, respectively. The existence of the constant is to avoid the situation where the denominator is 0. When $f_i > f_g$, there are natural enemies approaching in the surrounding environment, and when $f_i = f_g$, sparrows approach other individuals to avoid danger.

D. Improved Sparrow Search Algorithm

Aiming at the defects of SSA with poor population diversity and easy to fall into local optimum [30], this article proposes a multistrategy ISSA.

1) Optimizing the Diversity of Sparrow Population by Chaotic Mapping: Studies have shown that the initial population distribution of the bionic optimization algorithm has a significant impact on the optimization effect of the algorithm [29]. Namely, the more uniform the initial position distribution of the population leads to, the better the search speed and the optimization performance of the algorithm. And it can improve the diversity of the population without changing the randomness of the population by optimizing the initial position of the population with chaotic maps. The research shows that among the commonly used chaotic models, the Sin chaotic model has better chaotic characteristics than the logistic chaotic model because of its infinite number of mapping folds. The Sin chaotic model is adopted in this article to optimize the population diversity of the SSA, whose expression is shown as

$$\begin{cases} x_{n+1} = \sin\dfrac{2}{x_n}, & n = 0, 1, \ldots, N \\ -1 \leq x_n \leq 1, & x_n \neq 0. \end{cases} \tag{10}$$

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2) Improving SSA Search Strategy: According to the search strategy of the original SSA, its search strategy can be divided into two situations: approaching the origin and jumping to the position of the optimal solution. These two situations lead to the SSA easily falling into local optimum, and the global search ability being weak. To make up for the deficiency of SSA, this article makes improvements on the basis of the original SSA search strategy by removing the strategy of the sparrow approaching the origin and modifying the position update method of the finder as shown in the following equation:

$$x_{i,j}^{t+1} = \begin{cases} x_{i,j}^{t} \cdot (1+\delta), & R_2 < ST \\ x_{i,j}^{t} + \delta, & R_2 \geq ST \end{cases} \tag{11}$$

where δ is a random value that follows a normal distribution.

At the same time, the position update method of vigilantes is modified by the following strategy. When the sparrow approaches other individuals to avoid danger, if the sparrow's position is the optimal position of the current population, the sparrow will randomly move between the optimal position and the worst position of the current population. Otherwise, the sparrow moves randomly between the current position and the optimal position of the population. The improved position update method of vigilantes is shown in the following equation:

$$X_{i,j}^{t+1} = \begin{cases} X_{i,j}^{t} + \beta \cdot \left| X_{i,j}^{t} - X_{\text{best}}^{t} \right| & f_i > f_g \\ X_{i,j}^{t} + \beta \cdot \left| X_{\text{worst}}^{t} - X_{\text{best}}^{t} \right|, & f_i = f_g. \end{cases} \tag{12}$$

3) Introducing the Cauchy–Gaussian Mutation Strategy: In the later stage of the original SSA iteration, it is easy to fall into the local optimal situation. To alleviate this situation, this article adopts the Cauchy–Gaussian mutation strategy [30] to mutate the individual with the best fitness value in the current population. The Cauchy–Gaussian mutation strategy improves the global exploration capability, convergence accuracy, and stability of the algorithm by comparing the values before and after mutation position and selecting a better position for the next iteration. The specific formula of such improvement is as follows:

$$L_{\text{best}}^{t} = X_{\text{best}}^{t}\left[1 + \eta_1 \text{cauchy}\left(0, \sigma^2\right) + \eta_2 \text{gauss}\left(0, \sigma^2\right)\right] \tag{13}$$

$$\sigma = \begin{cases} 1, & f(X_{\text{best}}) < f(X_i) \\ \exp\left(\dfrac{f(X_{\text{best}}) - f(X_i)}{|f(X_{\text{best}})|}\right), & f(X_{\text{best}}) > f(X_i) \end{cases} \tag{14}$$

where L_{best}^{t} is the position of the optimal individual in the population after mutation; σ^2 is the standard deviation of the Cauchy–Gaussian mutation strategy; $\text{cauchy}(0, \sigma^2)$ and $\text{gauss}(0, \sigma^2)$ are random variables satisfying Cauchy and Gaussian distribution, respectively; and η_1 and η_2 are random numbers that obey the normal distribution.

With the aid of the global search advantage of the Cauchy mutation strategy and the local search advantage of the Gaussian mutation, the algorithm naturally jumps out of the local optimum and has a good ability of global search.

E. PF-ISSA Fusion-Based Positioning Algorithm

In many typical applications such as indoor inspection of mobile robots, the height of work scene usually has little changes, and its pitch angle does not swing violently. As a result, in this case the requirement of height information of the system is tiny, so this article only considers indoor positioning in a 2-D plane. The presence of obstacles renders most of the complex indoor environments being nonlinear, and the particle filter is a power tool for solving nonlinear and non-Gaussian dynamic model problems [18]. Therefore, the PF-ISSA algorithm is used to address the nonlinear problem of the combined positioning system of UWB and LiDAR.

In the combined positioning system, the time tags to UWB and LiDAR are first added by the computer for time synchronization. In addition, UWB and LiDAR are fixed on the combined platform so that the centers of these two sensors are at the same plumb line to achieve spatial synchronization. Then the displacement increment of the UWB ranging observations and the angle observations of the LiDAR SLAM is used as the measurement values. As a result, the combined UWB/LiDAR positioning model is established whose state equation is described as

$$X_t = f(X_{t-1}, W_{t-1}) \tag{15}$$

$$Z_t = g(X_t, V_t) \tag{16}$$

where $f(\cdot)$ is a state function based on time, and $g(\cdot)$ is a nonlinear measurement value in the sensor measurement data. W_{t-1} is the discrete white noise in the time domain, and V_t represents the measurement sequence of noise.

The measurement equations of the combined system are shown below

$$r_{t,i}^{U} = d_{t,i}^{U} + \xi_{i,t} = \sqrt{\left(x_t^U - x_{t,i}^B\right)^2 + \left(y_t^U - y_{t,i}^B\right)^2} + \xi_{i,t}^{U} \tag{17}$$

$$\Delta d_k^L = \sqrt{\left(x_t^L - x_{t-1}^L\right)^2 + \left(y_t^L - y_{t-1}^L\right)^2} + v_{\Delta d_k}^L \tag{18}$$

$$\alpha_t^L = \theta_1 + \alpha_1 - \gamma_1 + \gamma_2 = \theta_1 + \arctan\frac{\Delta x_{t-1}}{\Delta y_{t-1}} - \arctan\frac{d_{t-1,y}^L}{d_{t-1,x}^L} + \arctan\frac{d_{t,y}^L}{d_{t,x}^L} + v_{\alpha_k} \tag{19}$$

where (x_t^U, y_t^U) is the coordinate of UWB positioning at the moment t, $(x_{t,i}^B, y_{t,i}^B)$ is the coordinate of the ith UWB base station at the moment t, $d_{t,i}^U$ is the actual distance between the combined positioning system and the ith UWB base station at the moment t, $r_{t,i}^U$ is the measured value of the distance, Δd_t^L is the measured value of the displacement change in the LiDAR at adjacent moments, (x_t^L, y_t^L) and (x_{t-1}^L, y_{t-1}^L) are the coordinates of the LiDAR at moments t and $t-1$, respectively, α_t^L is the azimuthal angle of the LiDAR relative to the coordinates in the global coordinate system at moment t, Δx_{t-1} and Δy_{t-1} are the components of the UWB data changes in the x- and y-directions of the global coordinate system at time $t-1$ relative to time $t-2$, respectively, $d_{t-1,x}^L$ and $d_{t-1,y}^L$ are the components of the displacement changes in the x- and y-directions of the LiDAR coordinate system

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relative to the origin at time $t-1$ relative to time $t-2$, respectively, $d_{t,x}^{L}$ and $d_{t,y}^{L}$ are the components in the x- and y-directions of the change in displacement of the origin of the LiDAR coordinate system at moment t with respect to the origin of the coordinates at moment $t-1$, respectively, and $\xi_{i,t}^{U}$, $v_{\Delta d_t}^{L}$, and v_{α_t} are observation noise sequences.

The latest measurements of the combined system are used in the calculation of the fitness value of "sparrow," and the fitness function is

$$f(x_t^{(i)}) = \exp\left[-\frac{(z_t - x_t^p)^2}{2R}\right] \tag{20}$$

where R is the variance of the measurement noise, z_t is the latest measurement, and x^p is the particle prediction.

The flow of the PF-ISSA-based fusion positioning algorithm proposed in this article is as follows.

Step 1: Initialize the relevant parameters, and set the population size, dimension, maximum number of iterations, termination condition, and the proportion of detectors and vigilantes in the sparrow population. Sample N particles $\{x_0^i, i = 1, \ldots, N\}$ at the initial moment and obey the importance density function

$$x_t^i \sim \left(x_t^j | x_{t-1}^i, z_t\right) = p\left(x_t^j | x_{t-1}^j\right). \tag{21}$$

Step 2: Set the weight of each particle to $w_i = 1/N$.

Step 3: An ISSA is adopted to evolve the particles and the initial distribution of the sparrow population is optimized using Sin chaotic mapping according to (10).

Step 4: Update the current position of finder in the sparrow population according to the position iteration rule of (11), and subsequently the follower in the population moves in accordance with the position iteration rule of (8). In accordance with the above, the position of the vigilant in the population is updated according to formula (12).

Step 5: The particle with the best fitness value in the current iteration is perturbed with the Cauchy–Gaussian variant and its position is updated. Give the information of the particle after PF-ISSA resampling.

Step 6: Normalize with the weights

$$\tilde{w}_i(t) = \frac{w_i(t)}{\sum_{i=1}^{N} w_i(t)}. \tag{22}$$

Step 7: Output the information of particle states.

F. Optimizing the Global Pose Using Graph Optimization

Graph optimization means expressing the optimization problem in the form of a graph. The graph is composed of vertices and edges, which can be $G = \{V, E\}$ with V being the set of vertices and E being the set of edges. With the measured value of LiDAR and the measured value of UWB base station, the combination function is constructed as

$$e_t = z_t - h\left(x_t, x_i^m\right) \tag{23}$$

where z_t is the measured value of the combined system, x_t is the pose of the combined system at time t, x_i^m is the coordinates of the landmark, and e_t is the error amount.

Let $V = \{x_1, x_2, \ldots, x_t, x_1^m, x_2^m, \ldots, x_t^m\}$ be the set of combined system positions and poses and landmark positions. $E = \{z_1, z_2, \ldots, z_t\}$ is the set of measured values of the combined system by UWB at each moment, and $\{x_t, x_i^m\}$ is the optimization variable. Then, the objective function can be given in the following equation, where Ω_t is the inverse matrix of the covariance matrix:

$$\min_x \sum_{t=1}^{n} e_t^T\left(x_t, x_i^m, z_t\right)\Omega_t e_t\left(x_t, x_i^m, z_t\right). \tag{24}$$

Noting that z_t is known, formula (24) can be written as

$$\min F(x) = \sum_{t=1}^{n} e_t^T(x_t)\Omega_t e_t(x_t). \tag{25}$$

To obtain the optimized pose, one needs to give the starting point and the direction of iteration. First for the tth side $e_t x_t$, set the initial point obtained by moving prediction as $\tilde{X}_t$, and add an increment Δx. Then the estimated value of the side is $F_t(\tilde{X}_t + \Delta x)$, which means that the error is $e_t(\tilde{X}_t + \Delta x)$. With the first-order Taylor expansion, one has

$$e_t(X_t + \Delta x) \approx e_t(X_t) + \frac{de_t}{dx_t}\Delta x = e_t + J_t\Delta_t \tag{26}$$

where J_t is a comparable matrix.

For the objective function of the tth edge, one has

$$F_t(\tilde{X}_t + \Delta x) = e_t(\tilde{X}_t + \Delta x)^T \Omega_t e_t(\tilde{X}_t + \Delta x) \tag{27}$$

which together with the following equation yields:

$$F_t(\tilde{X}_t + \Delta x) \approx (e_t + J_t\Delta x)^T \Omega_t(e_t + J\Delta x) \tag{28}$$

that is,

$$F_t(\tilde{X}_t + \Delta x) = C_t + 2b_t\Delta x + \Delta x^T H_t \Delta x \tag{29}$$

where C_t is a constant term, $2b_t$ is a coefficient of a first-order term, and H_t is a coefficient of a quadratic term, namely, the Hessian matrix.

By increasing the increment, the change value of the objective function is given as

$$\Delta F_t = 2b_t\Delta x + \Delta x^T H_t \Delta x. \tag{30}$$

Since the goal of optimization is to make the increment being minimum, the derivative of ΔF_t about Δx is zero, that is,

$$\frac{dF_t}{d\Delta x} = 2b + 2H_t\Delta x = 0. \tag{31}$$

As a result, the following linear equation is obtained:

$$H_t\Delta x = -b_t \tag{32}$$

where $H = \sum J_t^T \Omega_t J_t$ and $b = \sum J_t^T \Omega_t e_t$.

Set the predicted coordinates of the movement of the combined system as the initial value, then the convergence direction can be obtained by solving the linear equation (29). For the obtained convergence direction, the iteration is carried out to ensure its convergence using optimization methods such as Gauss–Newton iteration. Finally, the optimal Δx is obtained, which is the pose of the combined system after global optimization.

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TABLE I
INFORMATION OF BENCHMARK FUNCTIONS

No.	Name	Dim	Scope	mini
f_1	Schwefel's Problem 2.22	30	[-100,100]	0
f_2	Schwefel's Problem 1.2	30	[-10,10]	0
f_3	Step Function	30	[-30,30]	0
f_4	Generalized Penalized Function 1	30	[-50,50]	0
f_5	Generalized Penalized Function 2	30	[-50,50]	0

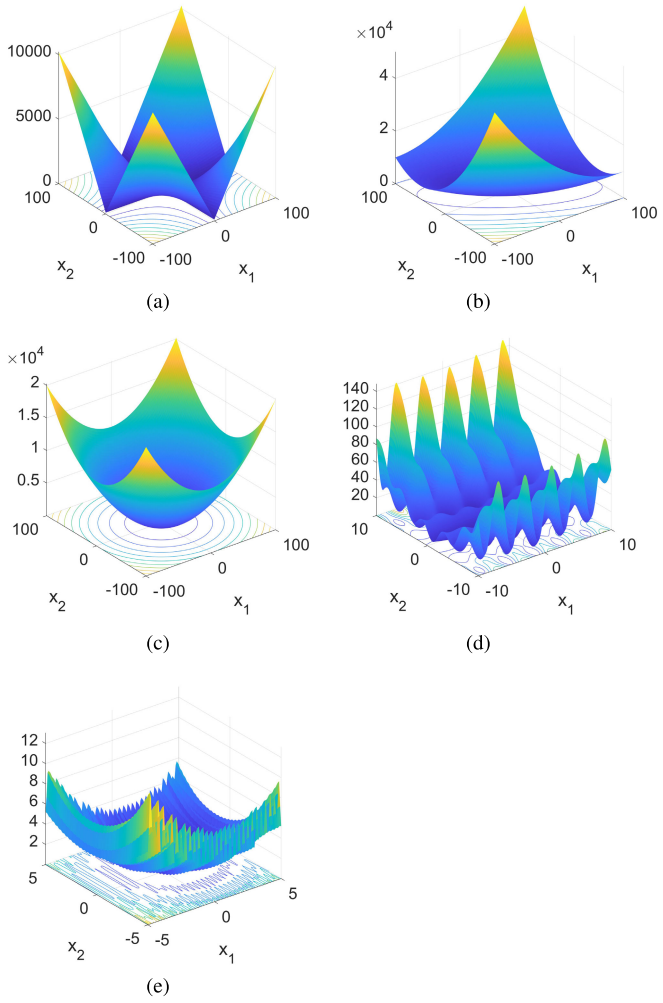


Fig. 5. Benchmark function image chosen for this article. (a) Schwefel's Problem 2.22. (b) Schwefel's Problem 1.2. (c) Step function. (d) Generalized penalized function 1. (e) Generalized penalized function 2.

III. RESULTS

A. Simulation Results

To verify the performance of ISSA given in this article, five commonly used benchmark functions are used in the simulation experiments, the information and images of which are given Table I and Fig. 5. The experiments are carried out under the environment of MATLAB R2022b, the operating system of 64-bit Win10, the memory of 32 GB, and the processor of AMD 5800X.

For the fairness of the experiments, the parameters of the test algorithms are set the same. That is, the number of SSA and ISSA populations is 50; the maximum number of iterations is 1000; the ratio of population discoverers is 0.2, and the ratio of vigilantes is 0.2. Particle swarm algorithm (PSO), bat algorithm (BA), and genetic algorithm (GA) are run simultaneously for the comparative experiments. In the PSO, c_1 and c_2 of the learning factors are set to 2, and the inertia weights ω are set to 1. In the BA, the loudness attenuation coefficient α is set to 0.85, the frequency attenuation coefficient β is set to 0.9, the frequency range is set to 0–1, and the loudness range is set to 1–2. In the GA, the crossover operator CR is set to 0.8, and the variance operator F is set to 0.05. The experimental data are recorded by running each benchmark function for 30 times, respectively. The mean value, standard deviation, and average running time of the obtained data results are as the evaluation indexes. The specific experimental data are shown in Table II, and the convergence curves of some of the test functions of the above algorithms are shown in Fig. 6.

It is clearly seen that compared with other optimization algorithms, the given ISSA has the advantages of faster convergence speed and higher accuracy. The simulation results show that ISSA can effectively overcomes the limitations of SSA such as poor global search ability and susceptible to the influence of local optimum, and thus, it can effectively improve the optimization ability of SSA.

In addition, the simulation localization experiments are also conducted to verify the performance of the PF-ISSA algorithm. These experiments test the positioning effects of the traditional PF with 50, 100, and 200 particles, as well as the PF-ISSA algorithm with 50 particles. Fig. 7 is given to depict the positioning effect of different algorithms.

The mean and root-mean-square errors of the fused trajectories and the errors of the real trajectory in the simulation experiments are presented in Table III below. The simulation data show that the PF-ISSA localization effect is similar to that using 200 particles in the PF, validating the algorithm's effectiveness.

B. Real Machine Experimental Results

To further validate the effectiveness of the proposed algorithm, a test scenario is set up in the experimental building's hall and a quadruped robot equipped with an NVIDIA TX2 running the Robot Operating System (ROS) operating system is chosen as the test platform. The UWB positioning module is the DW1000 series, while the LiDAR system used RPLIDAR-A2 manufactured by SLAM Technology. The UWB module typically achieves a positioning accuracy of 10 cm, with an effective communication range of 200 m in unobstructed conditions. The LiDAR measurement range spans from 0.2 to 8 m. The devices used in the experiment are depicted in Fig. 8.

Three UWB base stations are arranged in an indoor environment of about 50 m^2 with the randomly placed obstacles. And the robot is equipped with UWB positioning tags and LiDAR sensors. The setting of the experimental scene and the trajectories of the robot are shown in Fig. 9.

To guarantee the positioning accuracy, a laser rangefinder is used to measure the space size of the test scene and the position of the UWB base station. The measurement results are that the coordinate of base station 1 is (0, 0), the coordinate

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TABLE II
STATISTICS OF SIMULATION EXPERIMENTAL RESULTS

SSA			PSO			BA			GA			ISSA		
Mean	Rmse	Time	Mean	Rmse	Time	Mean	Rmse	Time	Mean	Rmse	Time	Mean	Rmse	Time
3.43E-28	4.12E-28	1.936	2.03E+00	2.56E-01	1.893	6.68E+01	7.08E+01	2.689	2.46E+01	6.49E+00	1.168	1.48E-34	1.81E-34	2.011
1.98E-27	2.25E-27	2.010	8.78E+02	2.42E+01	1.999	1.55E+04	4.83E+03	2.797	5.81E+03	1.48E+03	1.262	5.27E-32	8.65E-32	2.072
5.06E-02	2.01E-02	1.761	7.80E+01	2.18E+01	1.889	1.38E+04	3.91E+03	2.666	5.99E+03	1.52E+03	1.178	2.83E-05	1.23E-05	1.839
1.21E-2	4.58E-03	2.216	1.71E+00	5.69E-01	2.316	4.69E+07	3.49E+07	2.436	7.97E+06	7.18E-01	1.436	5.28E-6	2.25E-6	2.287
5.23E-02	1.44E-02	2.227	2.78E+00	6.70E-01	2.298	1.14E+08	6.46E+07	3.312	2.42E+07	1.99E+07	1.457	7.20E-05	5.19E-05	2.273

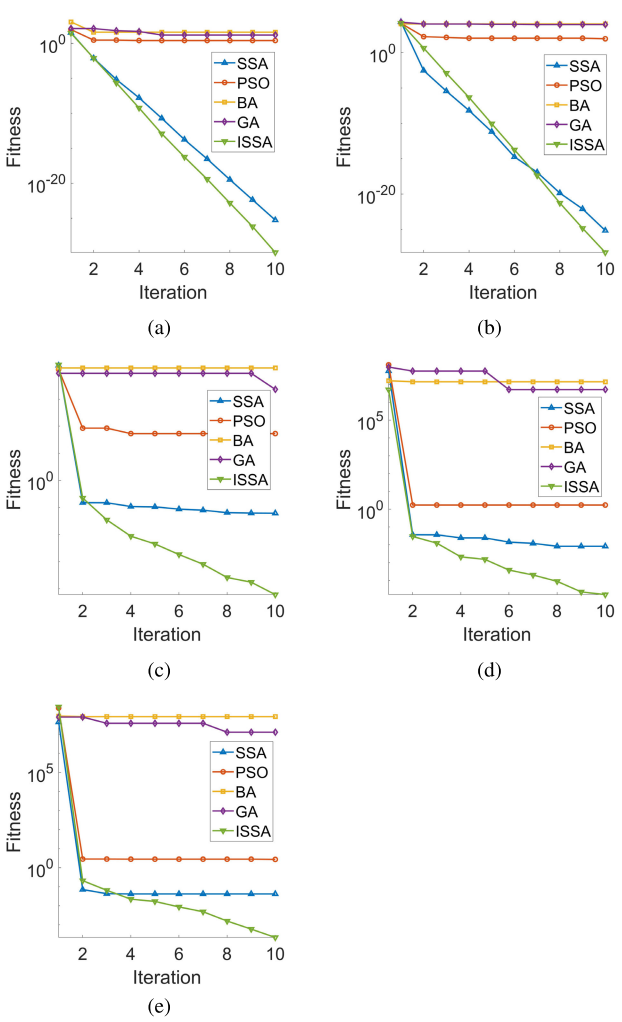


Fig. 6. Partial convergence curves of the benchmark functions in this article. (a) f_1. (b) f_2. (c) f_3. (d) f_4. (e) f_5.

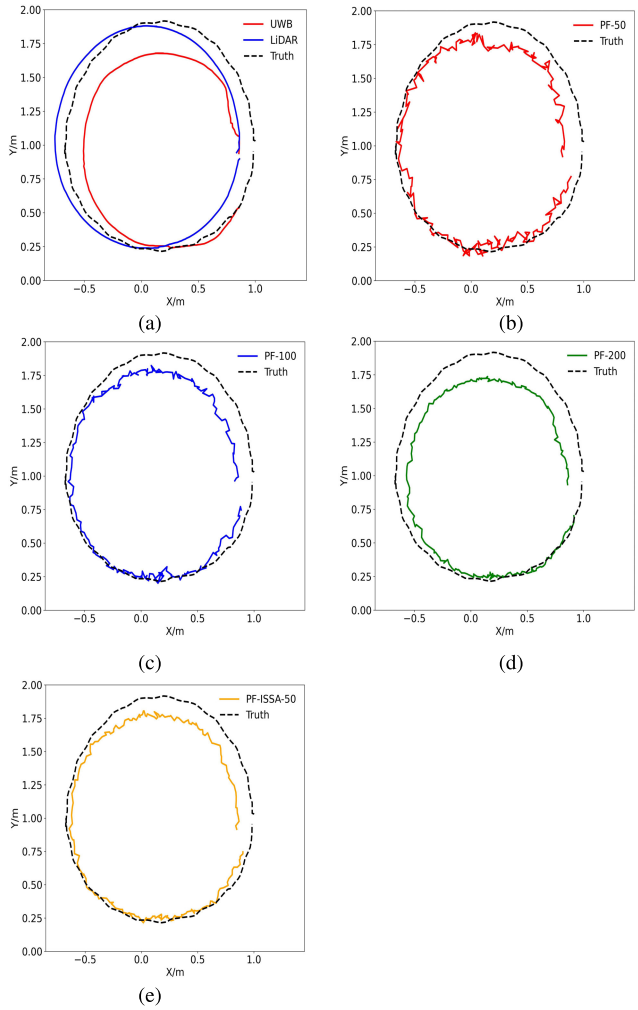


Fig. 7. Simulation of positioning results with different algorithms. (a) Individual positioning. (b) PF-50. (c) PF-100. (d) PF-200. (e) PF-ISSA-50.

of base station 2 is (0, 8), and the coordinate of base station 3 is (8, 0) (coordinate units are in meters). The movement path of the robot is carried out in a counterclockwise direction and the route is a square of 5 × 5 m. With this route, the following experiments are carried out in this article.

Experiment 1: The UWB positioning device is used alone for the experiment. The LiDAR sensor is used for standalone positioning experiments. The experimental results are presented in Figs. 10 and 11.

The results show that when UWB positioning is used alone, the mean value of positioning error is 0.392 m and the root-mean-square error of positioning is 0.428 m. It is worth noting that when there are obstacles blocking the UWB signals

TABLE III
DATA OF SIMULATED POSITIONING EXPERIMENT

Error/m	PF-50	PF-100	PF-200	PF-ISSA-50
Mean	0.161	0.135	0.129	0.130
Rmse	0.168	0.142	0.134	0.136

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Fig. 8. (a) UWB positioning module. (b) LiDAR.

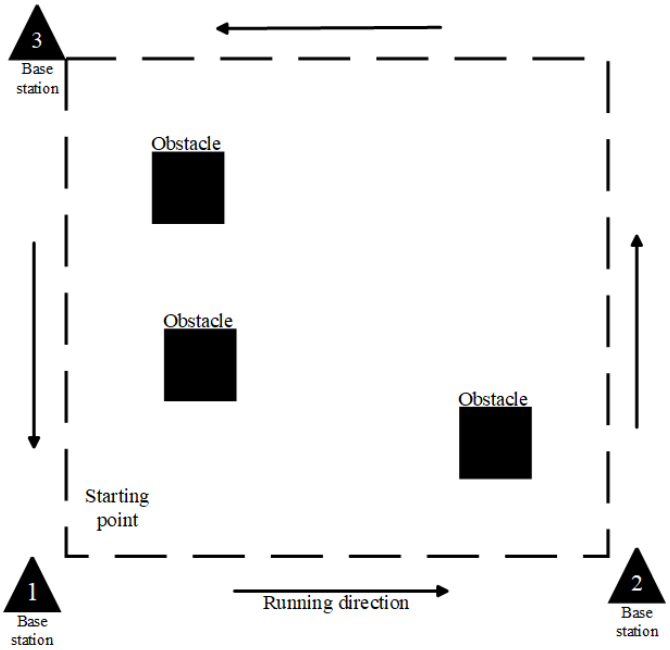


Fig. 9. Schematic of the test environment.

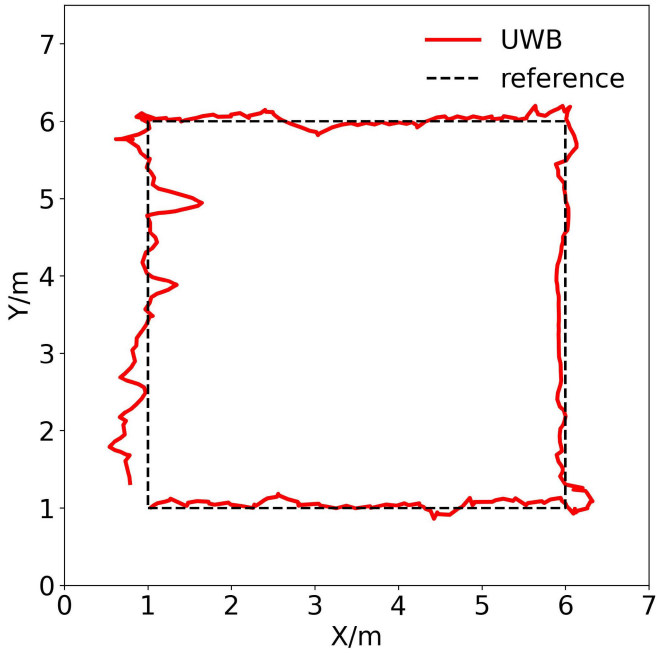


Fig. 10. UWB positioning track.

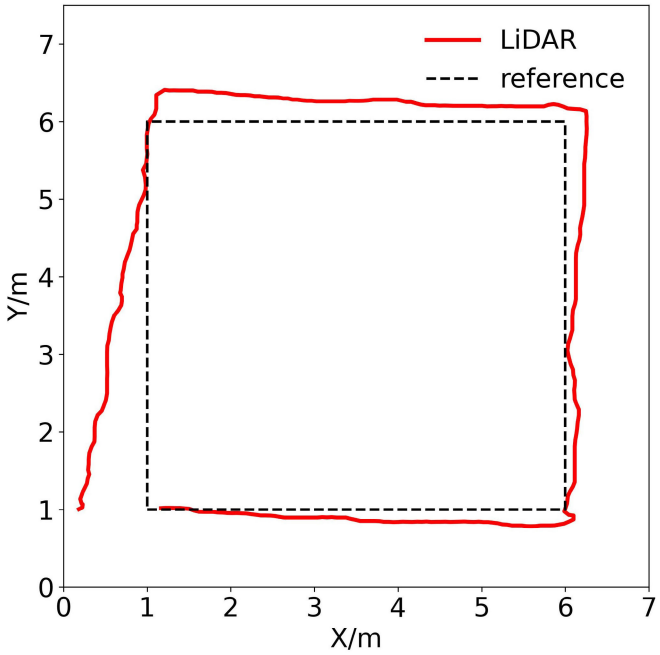


Fig. 11. LiDAR positioning track.

on the test routes from base station 1 to base station 2 and from base station 3 to base station 1, the resultant NLOS error affects the accuracy and stability of UWB positioning. When LiDAR alone is used for localization, the mean value of the localization error is 0.297 m, and the root-mean-square error of localization is 0.327 m. As seen from the figure, the use of a single LiDAR sensor for localization leads to significant cumulative errors, which adversely affects the accuracy of the device's localization.

Clearly, the above experimental results indicate that the positioning accuracy of a single sensor hardly meets the requirement of high-precision positioning.

Experiment 2: In this experiment, a combined positioning experiment using UWB and LiDAR is conducted. The positioning performance of the UWB/LiDAR system is tested using both the traditional particle filter algorithm and the particle filter algorithm optimized by ISSA. Specifically, the experiment is evaluated the positioning performance of the traditional particle filter algorithm with 50 particles, 100 particles, and 200 particles, as well as the particle filter algorithm after ISSA optimization with 50 particles. The experimental results are depicted in Fig. 12. And the maximum error and root-mean-square error of different particle number positioning trajectories and reference trajectories in Experiment 2 are listed in Table IV.

It is apparent from them that the positioning effectiveness of the particle filter algorithm is improved as the number of particles increases. The particle filter algorithm optimized by ISSA demonstrates comparable performance to the traditional particle filter algorithm with 200 particles, but with only 50 particles. This reduction in the number of particles amounts to a 75% decrease, effectively reducing the algorithm's complexity.

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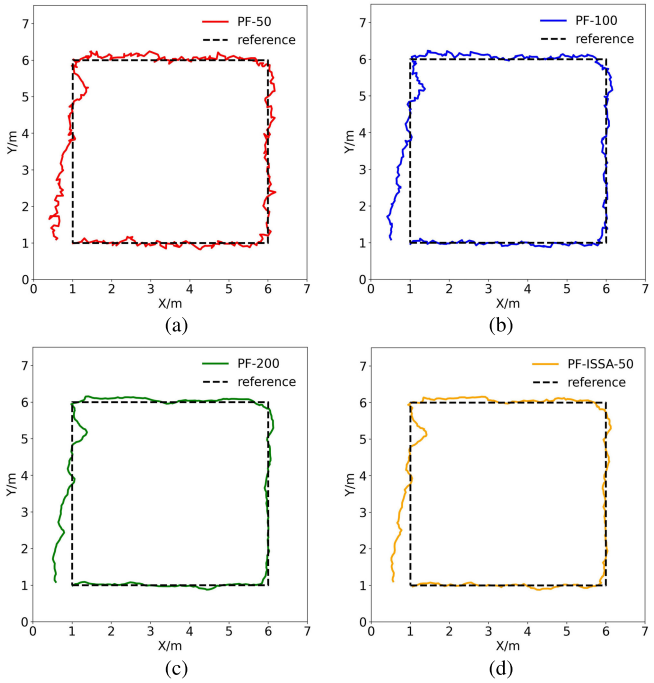


Fig. 12. Positioning effect of the particle filter algorithm using different particles. (a) PF-50. (b) PF-100. (c) PF-200. (d) PF-ISSA-50.

TABLE IV
TRAJECTORY ERRORS OF POSITIONING EXPERIMENT

Error/m	PF-50	PF-100	PF-200	PF-ISSA-50
Mean	0.432	0.307	0.244	0.253
Rmse	0.502	0.351	0.292	0.295

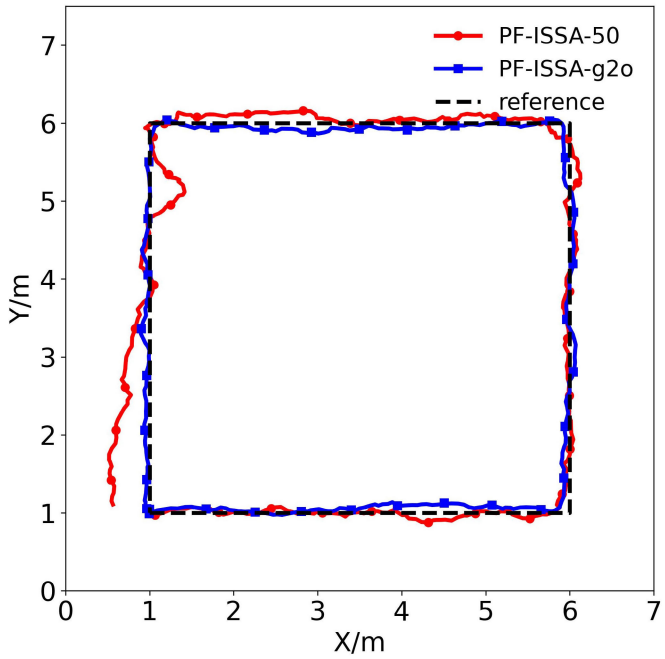


Fig. 13. Comparison of positioning effects with and without graph optimization.

Experiment 3: On the basis of experiment 2, the back-end graph optimization is further added to the UWB/LiDAR combined positioning system. The positioning effects with and without such graph optimization are given in Fig. 13. Furthermore, to demonstrate the superiority of the proposed algorithm, the performance comparison experiment of our algorithm and the EKF-based fusion positioning algorithm in [15] is also implemented, whose results are shown in Fig. 14.

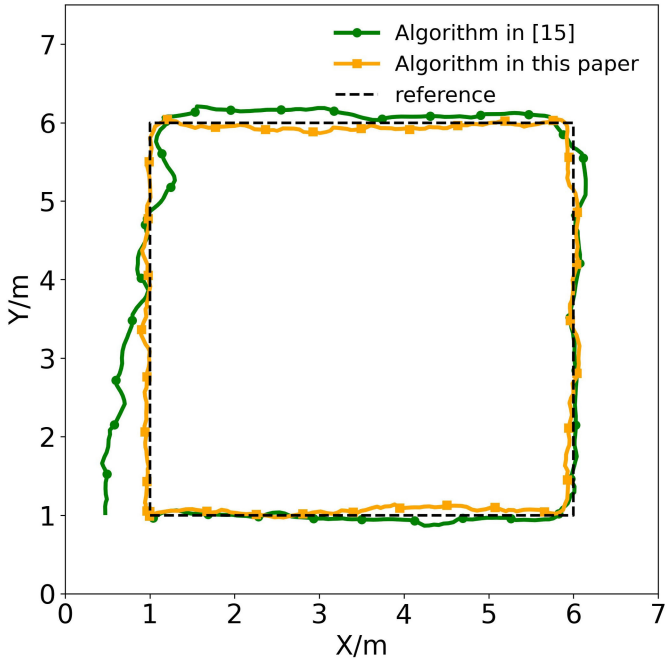


Fig. 14. Comparison between the proposed positioning algorithm and the EKF-based positioning algorithm in [15].

Fig. 15 illustrates the progression of positioning error over time using UWB, LiDAR, EKF-based fusion positioning, and the algorithm proposed in this study. It is evident that UWB positioning results in abrupt changes in positioning error when the obstacles are encountered, while the LiDAR positioning error steadily accumulates and gradually increases. Comparatively, the positioning error of the algorithm proposed in this article is smaller than those of UWB, LiDAR, and the EKF-based algorithm in [15]. Therefore, it can be concluded that the algorithm proposed in this article exhibits higher accuracy and robustness.

From the quantitative comparison given in Table V, it is evident that the UWB/LiDAR tightly coupled positioning method proposed in this article demonstrates a substantial increase in positioning accuracy, which presents a respective improvement of 69.16% and 59.63% when compared with single-sensor positioning. In addition, in comparison to the EKF-based fusion positioning [15], the proposed method shows an improvement in positioning accuracy by 55.71%. The incorporation of tight coupling at the front end effectively enhances the local positioning accuracy of the robot and mitigates the impact of NLOS errors on UWB positioning accuracy. Furthermore, graph optimization at the back end successfully alleviates cumulative errors generated by LiDAR positioning, thereby leading to enhanced positioning accuracy and system robustness.

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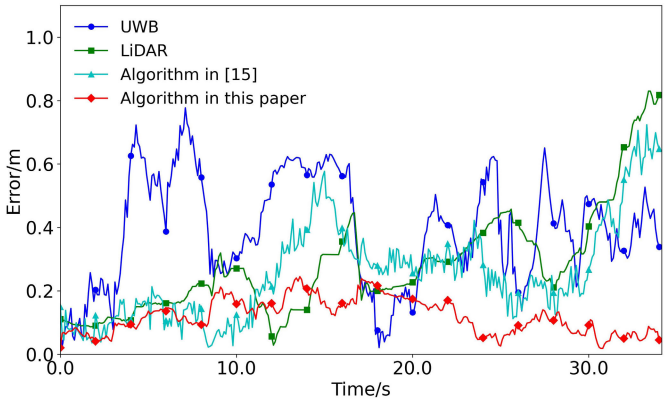


Fig. 15. Curves of error versus time for different positioning algorithms.

TABLE V
POSITIONING ERRORS OF DIFFERENT POSITIONING ALGORITHMS

Error/m	UWB	LiDAR	Algorithm in [15]	Algorithm in this paper
Max	0.392	0.282	0.255	0.120
Rmse	0.428	0.327	0.298	0.132

IV. DISCUSSION

In indoor environments where the area is not too large, the proposed method requires a lower cost of sensors compared with the 3-D LIDAR with higher accuracy. Meanwhile, the implementation of this method is also relatively easy. The LiDAR measurement information is used to offset the NLOS errors in the UWB measurement information. Both the sensors constrained each other, so the combination of the front-end filter and the back-end optimization can achieve high-precision indoor positioning. However, in the actual arrangement of UWB positioning base stations, the placement of base stations should be concerned about. Especially, positioning base stations should be placed in a square position, and the distance between each positioning base station is normally less than 40 m. When the applications in a wide range of scenarios are conducted, there are certain requirements for the number of UWB positioning base stations. The multiple UWB positioning base stations may lead to an increase in the cost.

V. CONCLUSION

Based on the improved particle filter, this article has proposed a UWB/LiDAR indoor positioning method using the respective advantages of these two sensors for fusion positioning, that is, the measured value of LiDAR is used to offset the impact of the NLOS error in UWB positioning. An ISSA is introduced into particle filter to effectively eliminate the particle degradation phenomenon and improve the algorithm speed and real-time performance with a small number of particles. The experimental results show that compared with the single-sensor positioning method of UWB or LiDAR, the proposed UWB/LiDAR fusion positioning method can achieve significant improvements in positioning accuracy with enhancing 69.16% and 59.63%. In addition, the positioning accuracy of the proposed method is also improved 55.71% compared with EKF-based fusion positioning. It indicates that the proposed method effectively addresses the limitations and mitigates the errors associated with individual sensors and results in more precise and reliable positioning outcomes.

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